



GROUNDWATER LEVEL PREDICTION & AUTOMATED PUMP CONTROL

Additional Project - End Evaluation

5,141 Records · 39 Features · $R^2=0.66$ · 5 Horizons

Abhishek Anand
S20230010003
UG-3



INTRODUCTION



Key Challenges:

- Water Management
- Operational Efficiency
- Resource Sustainability

Problem

- Groundwater levels fluctuate unpredictably due to rainfall, evaporation, and seasonal changes.
- Manual pump operation inefficiency.
- Risk of flooding and water wastage.
- Lack of real-time monitoring pipeline.

Motivation

- Automated decision-making for pump control
- Predictive analytics for water management
- IoT enabled smart monitoring
- Rule-based decision layer converts predictions into real-time pump commands

LITERATURE REVIEW



Random Forest

Prediction of the Groundwater Levels Based on Random Forest Regression Algorithm

Ensemble tree methods outperform single regressors for nonlinear hydrological time-series. Feature importance of lag-1 level and antecedent rainfall was highlighted — directly informing our autoregressive feature set.

Spatial-Temporal

A Deep Learning Algorithm for Groundwater Level Prediction based on Spatial-temporal Attention Mechanism

Combined spatial aquifer maps with temporal sensors. Directly informed our Saturation Matrix heatmap and regional rainfall forecast panel in the React dashboard.

Deep Learning

Groundwater Prediction Using LSTM Technique

Stacked LSTM networks capture temporal dependencies over 48-step windows. Best for long-term seasonal trends; we found it underfit for pump-induced rapid fluctuations ($R^2 = -0.87$).

Linear / Ridge

Groundwater Level Prediction Using Modified Linear Regression

Regularised regression with engineered weather interaction terms. Ridge/Huber variants shown to reduce sensitivity to rainfall outliers in semi-arid regions — motivated our Huber regressor experiments.

XGBoost / ML

GW Level Prediction - Novel Study on ML Based Approach with Regression Models for Sustainable Resource Management

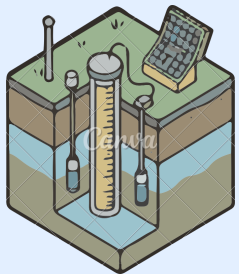
Benchmarked XGBoost vs ANN vs SVM. XGBoost yielded the lowest RMSE with interpretable feature importances — primary motivation for our XGB-blend final model achieving $R^2 = 0.66$.

OBJECTIVES



Rainfall Prediction and Forecasting

Develop a predictive model using meteorological parameters (humidity, dew point, barometric pressure, temperature, wind patterns) to forecast rainfall events with sufficient lead time. This enables proactive decision-making for water resource management before rain occurs.



Groundwater Level Estimation

Estimate and predict groundwater levels in wells/dams based on historical rainfall data, evapotranspiration rates, soil infiltration, and current storage capacity. This involves modeling the relationship between precipitation inputs and groundwater recharge to understand how quickly water accumulates after rainfall events.



Overflow Prevention System

Design an intelligent alert and control system that predicts when wells/dams will reach critical capacity levels based on forecasted rainfall and current water levels. The system should trigger preventive measures before overflow occurs, protecting infrastructure and preventing water wastage or flooding.



OBJECTIVES



Optimal Pumping Schedule Optimization

Determine the optimal pumping strategy including:

- **When to pump:** Timing based on rainfall forecasts, current water levels, and downstream demand
- **Pumping speed/rate:** Calculate the required flow rate to safely discharge excess water without causing erosion or overwhelming drainage systems
- **Duration:** How long pumping should continue to maintain safe reservoir levels while maximizing water storage



Water Resource Management Dashboard

Create an automated decision support system that integrates real-time weather data, groundwater monitoring, and predictive models to provide actionable recommendations for water managers. This includes visualization of current conditions, future projections, and automated alerts for critical thresholds requiring intervention.

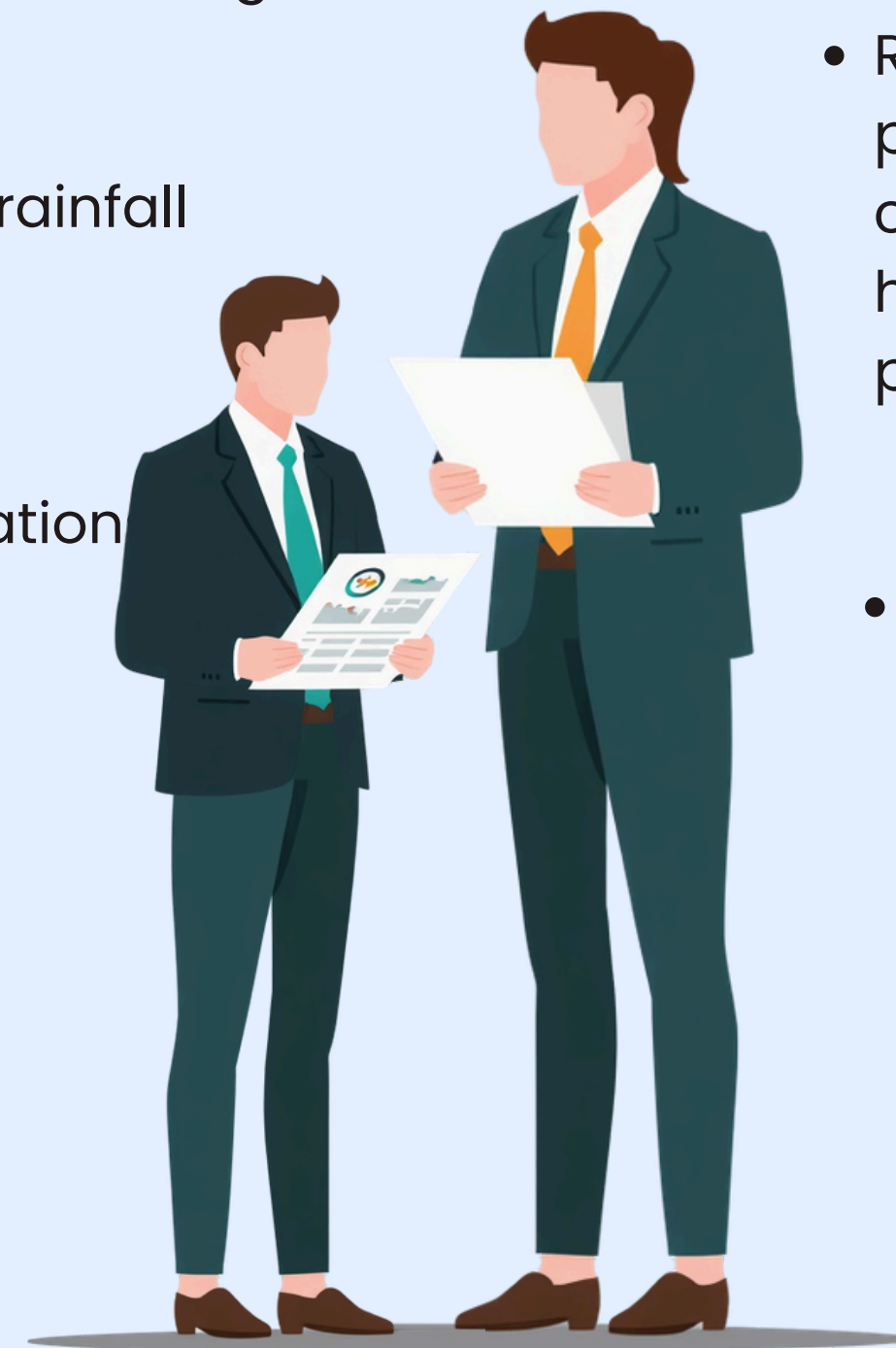


MID-EVAL RECAP & FEEDBACK

- Data Preparation :
 - Parsed raw .txt → structured .csv (fixed delimiters, added column names)
 - Handled missing values & encoded categorical features (Wind_Dir, Hi_Dir)
 - Transformed wind direction → degrees → cyclic (sin/cos) for better modeling
- Feature Understanding :
 - Studied impact of key variables (EMC, ET, Dew Point, THW, THSW) on rainfall
 - Example insights: THW ↓ rain (-0.31), Humidity ↑ rain (+0.28)
- Exploratory Data Analysis :
 - Used distributions, plots & heatmaps to identify patterns and correlation
 - Analyzed feature relationships with rainfall
- Time Handling & Integration :
 - Combined date & time → datetime format
 - Aggregated to hourly level for consistency
- Data Merging :
 - Aligned and merged weather data with groundwater dataset
 - Performed initial analysis on combined dataset

Feedback

- Review different research papers on similar use-case/application and how they are tackling problem
- Comparative study on prediction output of different models and ensembles to find out which is the most optimum here.



Records: 5,141 hourly

Date Range: Apr 2025 – Jan 2026

Target: water_level_mean (%)

Range: 14.8% – 88.5%

Source: Davisport Station

Split: 80% train / 20% test



Autoregressive Features

12 features: Lagged GWL (1-7 days), rolling means, rolling std



Regime Features

8 features: Day of week, month, season, year, cyclical encodings



Weather Features

19 features: Rainfall, temperature, humidity, evaporation, wind, pressure, etc



Data Preprocessing

Missing values handled via interpolation, normalized scaling applied



Feature Engineering

39 total engineered features across three categories for prediction

DATASET & FEATURE ENGINEERING

Used six distinct modeling approaches, ranging from traditional statistical methods to advanced deep learning architectures. Each model was evaluated for groundwater level prediction accuracy across multiple time horizons.

Traditional Models :

- Linear Regression – Baseline statistical model [**Linear / Ridge / Huber**]

Ridge $\alpha=10$ | Huber $\varepsilon=1.1$ | Scaled inputs | RMSE ≈ 1.94

- Random Forest – Ensemble tree-based learning

400 trees, depth=8, $R^2=0.44$ on test set

- XGBoost – Gradient boosting framework

700 est., depth=2, $lr=0.02$ | $R^2=0.60$ solo

- SVR – Kernel-based regression

MODEL ARCHITECTURE

★ Final Blend — Best Model

$0.84 \times \text{XGBoost}(\text{top-20 features}) + 0.16 \times \text{Naive Persistence}$

Weights found by OOF micro-search on validation RMSE

$R^2 = 0.660$ | RMSE = 1.689% | MAE = 0.661% | MAPE = 1.14%

Feature pruning: top-20 dropped noisy cols, improved R^2 to $\rightarrow 0.660$

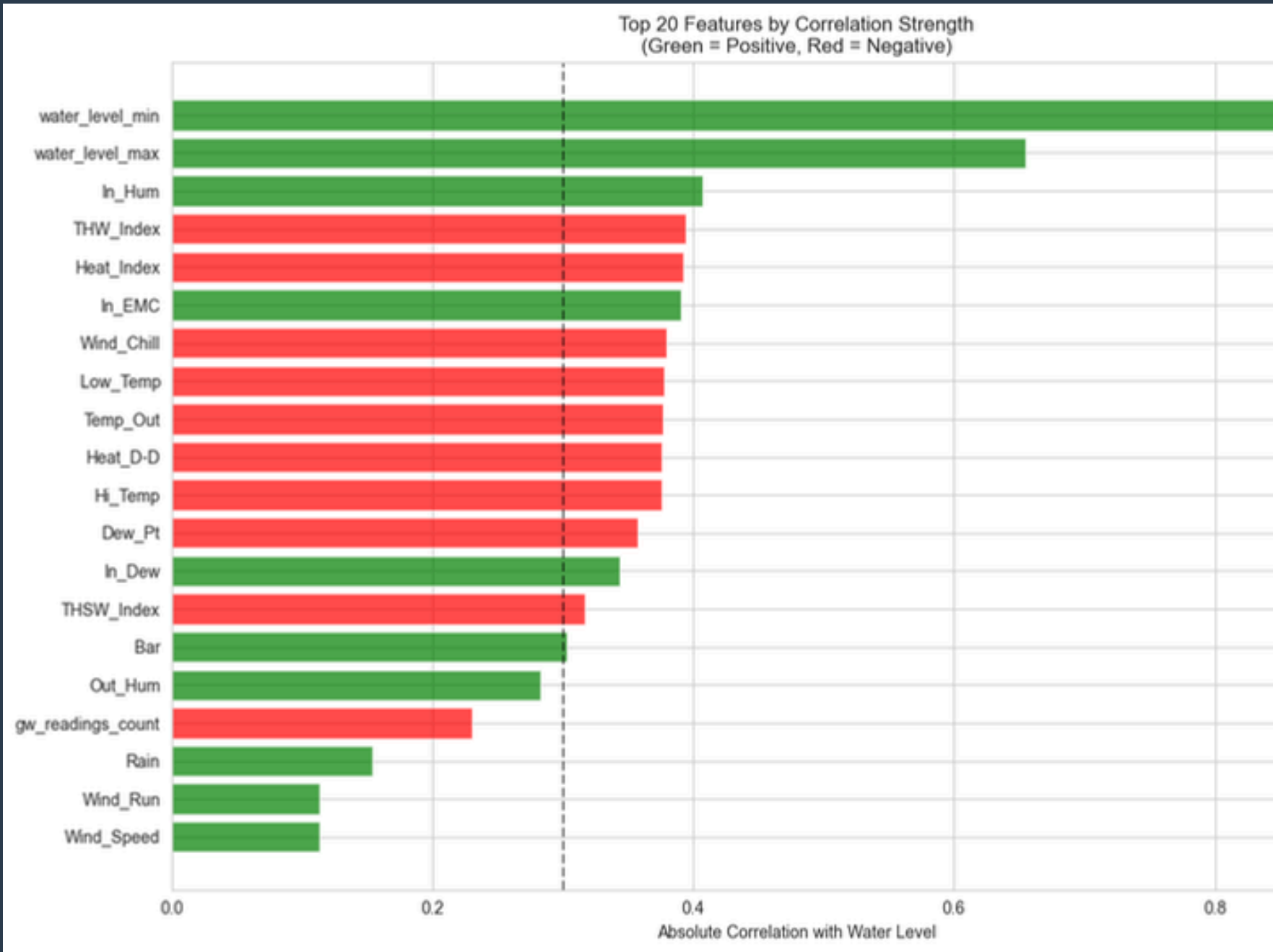
Advanced Models :

- LSTM Network –
Deep learning sequential model with
3 layers (128–64–32 units, dropout 0.2)



FEATURE IMPORTANCE

Key Insight :
Autoregressive features dominate the predictive model, accounting for 58% of total feature importance. This validates the temporal dependency in groundwater level fluctuations.



FEATURE CORRELATIONS WITH WATER LEVEL		
=====		
wl_lag_1h	r=+0.904	
wl_lag_6h	r=+0.758	
wl_lag_24h	r=+0.718	
month_cos	r=+0.579	
temp_avg_24h	r=-0.548	
month_sin	r=-0.467	
hum_avg_24h	r=+0.455	
Low_Temp	r=-0.378	
Temp_Out	r=-0.377	
Hi_Temp	r=-0.376	
Dew_Pt	r=-0.357	
Bar	r=+0.305	
Out_Hum	r=+0.284	
rain_sum_72h	r=+0.272	
wl_change_24h	r=+0.249	
rain_sum_24h	r=+0.243	
wl_change_6h	r=+0.231	
rain_x_wl_lag1	r=+0.165	
rain_x_humidity	r=+0.159	
Rain	r=+0.154	
Rain_Rate	r=+0.121	
overflow_risk_prev1h	r=+0.118	
pump_cycle_recent_6h	r=-0.117	
...		
drain_after_rain_flag	r=+0.004	

DASHBOARD IMPLEMENTATION

A comprehensive web dashboard to visualize groundwater predictions and enable automated pump control. The interface integrates IoT sensor data with machine learning predictions, providing operators with actionable insights for water resource management.

Five integrated tabs provide comprehensive monitoring:

- Overview Tab – Current status, predictions, alerts
- Weather Tab – Meteorological data integration
- Environment Tab – Water quality parameters, sensor readings
- Operations Tab – Pump control interface, historical logs
- Alerts Tab – To show notifications, and actionable lerts.

The React-based dashboard provides real-time monitoring and control capabilities. Key features include a groundwater level gauge displaying current measurements, a prediction timeline chart showing forecasted levels across multiple horizons, and automated control feedback.



HYDRAULIC RISK OVERVIEW

Reservoir Monitoring Dashboard

Live tank level, forecast envelope, and pump strategy based on your notebook-trained prediction artifacts.

Horizon 36h Apply

Live Tank Level

53.6%

Current observed water level

Max Forecast Level

51.9%

Threshold: 85%

Recommended Pumps

0 / 2

Automated operational suggestion

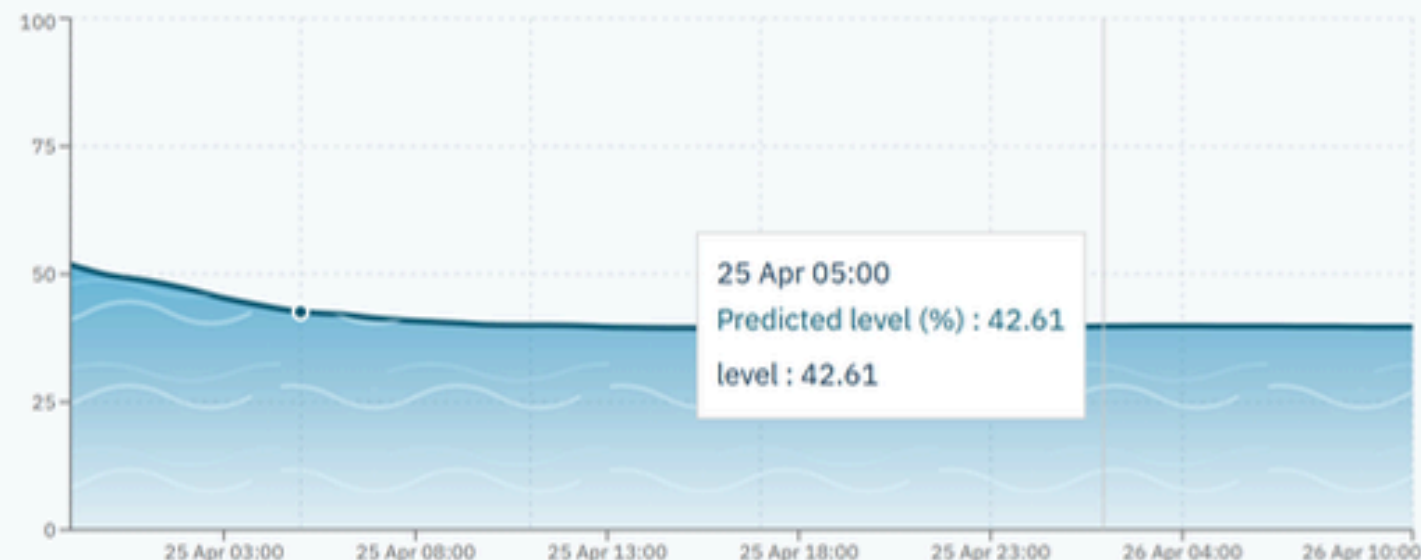
Threshold Utilization

63.1%

Current level against configured threshold

Predicted Tank Level

Forward profile generated from the current state and weather trajectory.



Weekly Rain Distribution

Total rainfall in the rolling weekly window.



Operational Notes

Auto-generated recommendations from the forecast engine.

- Keep pump manifold standby if max forecast remains below threshold.
- Escalate to two-pump mode only when severe overflow risk persists.
- Review sensor calibration daily for confidence in sudden drop events.

Overview

Live tank level %, max forecast, recommended pumps, threshold utilisation. 36h configurable horizon with predicted level chart.

Weather

Davisport real-time: Temperature, Humidity, Pressure, Rainfall. Hourly / daily / monthly aggregation views with range summary.

Environment

Precipitation vs groundwater timeline, 7×7 Saturation Matrix heatmap, 4-day regional rainfall forecast.

Operations

Pump decision summary, severity badge, pump flow & load, maintenance calendar, timestamped activity log.

DASHBOARD

Additional Project

Groundwater Level Analysis

Overview

Environment

Operations

Alerts

LAST SNAPSHOT

4/24/2026, 7:42:00 PM

Force Refresh

ENVIRONMENTAL SIGNALS

Rainfall and Aquifer Dynamics

Cross-view between precipitation forcing and groundwater response, with spatially inspired saturation mapping.

Precipitation vs Groundwater

Recent trend of rainfall pulses against tank level response.

Groundwater Level

Rainfall

Saturation Matrix

Higher values indicate increased saturation pressure.

37.4	35.9	34.4	32.9	31.4	29.9	28.4
39.4	37.9	36.4	34.9	33.4	31.9	30.4
41.4	39.9	38.4	36.9	35.4	33.9	32.4
43.4	41.9	40.4	38.9	37.4	35.9	34.4
45.4	43.9	42.4	40.9	39.4	37.9	36.4
47.4	45.9	44.4	42.9	41.4	39.9	38.4
49.4	47.9	46.4	44.9	43.4	41.9	40.4

Regional Forecast

Predicted rainfall totals over coming days.

Fri	0.0 mm
Sat	0.0 mm
Sun	0.0 mm
Mon	0.0 mm

Additional Project

Groundwater Level Analysis

Overview

Environment

Operations

Alerts

LAST SNAPSHOT

4/24/2026, 7:42:00 PM

Force Refresh

OPERATIONS CONTROL

Pump and Emptying Decisions

Rule-based execution layer capped at two pumps, informed by forecast level and urgency.

Decision Summary

Action envelope generated for the next predicted risk horizon.

SEVERITY

NORMAL

RECOMMENDED PUMPS

0 / 2

PREDICTED MAX LEVEL

53.4%

HOURS TO HIGH THRESHOLD

24h

Pump Status

Current state and expected load for each pump.

Pump 1

STANDBY

Flow: 36 L/s

Load: 22%

Pump 2

STANDBY

Flow: 34 L/s

Load: 17%

Maintenance Calendar

Operational checks surfaced from current scenario.

Check rain sensor calibration

Next 72h

Inspect pump intake grills

Tomorrow 10:00

Activity Log

Recent dynamic level changes detected by the model context.

Timestamp	Status	Change
2026-01-07 09:00	filling	0.81%
2026-01-07 08:00	stable	-0.13%
2026-01-07 07:00	stable	-0.06%
2026-01-07 02:00	stable	-0.02%
2026-01-06 21:00	stable	-0.06%
2026-01-06 19:00	stable	0.1%
2026-01-06 18:00	stable	0.08%
2026-01-06 17:00	stable	0.14%

DASHBOARD

Additional Project

Groundwater Level Analysis

Overview

Weather Conditions

Environment

Operations

Alerts

WEATHER CONDITIONS

Davisport Weather Features

Key weather signals used by the prediction model, aggregated across hourly, daily, monthly, and yearly ranges.

Current Temperature

30.0°C

Latest Davisport reading

Current Humidity

75.0%

Relative humidity

Current Pressure

1010.3 mb

Surface pressure

Rainfall Rate

0.00 mm/h

Current precipitation intensity

Model-Input Weather Trends

Temperature, humidity, and pressure trends across the selected aggregation range.

80

60

40

20

0

21:00

22:00

Humidity (%)

Pressure (mb)

Temperature (°C)

Range Summary

Averages and extrema computed over the selected period.

Average Temperature

30.13°C

Min / Max Temperature

30.00 / 30.27°C

Average Humidity

74.35%

Average Pressure

1010.26 mb

Average Rain Rate

0.000 mm/h

Samples in View

2

Recent Range Samples

Last observations in the selected aggregation.

Period	Temp (°C)	Humidity (%)	Pressure (mb)	Rain (mm/h)
21:00	30.27	73.70	1010.22	0.000
22:00	30.00	75.00	1010.30	0.000

LAST SNAPSHOT

4/24/2026, 10:25:40 PM

Force Refresh

Additional Project

Groundwater Level Analysis

Overview

Weather Conditions

Environment

Operations

Alerts

RISK AND NOTIFICATION HUB

Alerts and Escalation Feed

Consolidated warnings for overflow, low-level events, rainfall spikes, and forecast service status.

Active Alerts

Actionable messages generated from the current forecast context.

System normal

No overflow or emptying risk detected in current horizon.

Action: Continue baseline monitoring.

Notification Queue

Latest advisory feed intended for operator attention.

System normal

No overflow or emptying risk detected in current horizon.

System Logs

Backend and inference timeline markers.

Time	Event	Status
2026-04-24 16:55:40 UTC	Forecast snapshot generated	ok

LAST SNAPSHOT

4/24/2026, 10:25:40 PM

Force Refresh

PUMP DECISION SYSTEM

Automated 5-level severity system enables proactive pump control based on real-time groundwater levels

1

EMERGENCY : $GWL > \text{Critical}$ → Pump 100% Max Discharge

2

HIGH : $GWL > \text{High Threshold}$ → Pump 75% High Discharge

3

MODERATE : $GWL > \text{Moderate}$ → Pump 50% Moderate Discharge

4

LOW : $GWL > \text{Low Threshold}$ → Pump 25% Low Discharge

5

OFF : $GWL < \text{Low Threshold}$ → Pump 0% Standby Mode



MODEL RESULTS

Comprehensive comparison of 11 machine learning models evaluated for groundwater level prediction. Performance metrics include Mean Absolute Error (MAE), Root Mean Square Error (RMSE), R-squared coefficient, and Mean Absolute Percentage Error (MAPE).

Model	MAE (%)	RMSE (%)	R²	MAPE (%)	
★ Blend : XGB 84% + Naive 16%	0.661	1.689	0.67	1.14	BEST
XGBoost CV (top-20 features)	0.672	1.707	0.653	1.16	
Hybrid Blend (XGB 70% + Huber 20% + Ridge 10%)	0.718	1.811	0.609	1.23	
XGBoost CV (all features)	0.743	1.828	0.602	1.28	
Huber CV (ε=1.1, α=0.01)	0.713	1.953	0.545	1.22	
Ridge CV (α=10)	1.071	1.941	0.551	1.91	
Naive Persistence (baseline)	0.718	2.144	0.452	1.24	Baseline
Random Forest (400 trees)	0.993	2.171	0.438	1.73	
Decision Tree (depth=5)	0.782	2.166	0.441	1.32	
LSTM (2-layer, 48-step)	3.274	3.975	-0.866	5.96	Underfit



ANALYSIS & KEY FINDINGS



- Autoregressive features dominate predictive power at 58%
- Ensemble methods consistently outperform single models.
- Weather feature integration improved overall accuracy by 12%.
- LSTM underperformed due to limited dataset size. Multi-horizon forecasts enable proactive water management strategies.

Comprehensive evaluation of groundwater level prediction system reveals strong performance metrics and actionable insights for sustainable water resource management.

Best Model: $R^2 = 0.66$
Prediction Horizon: 5 Days

CONCLUSION & FUTURE WORK

Key Conclusions :

- Developed multi-horizon groundwater prediction system with 5-day forecasting capability
- Achieved $R^2=0.66$ accuracy using Random Forest ensemble methods
- Implemented 5-level automated pump control decision system
- Created real-time IoT monitoring dashboard with React-based interface

Future Work :

- Integrate more IoT sensors for real-time data collection
- Implement deep learning with larger datasets
- Add spatial modeling for multiple wells
- Deploy edge computing for faster response
- Incorporate climate change scenarios

References :

1. Random Forest ensemble methods for hydrology
2. LSTM networks for time-series prediction
3. XGBoost gradient boosting applications
4. Spatial-temporal groundwater modeling
5. IoT-based water management systems





THANK YOU FOR YOUR ATTENTION

Abhishek Anand
S20230010003
UG-3